Satyadarshi Das

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.32
ST Xaviers High School ,Kedar Gouri , Bhubaneswar	Class XII (Central board of Secondary Education)	2019	86.20%
DAV Public School ,Kalinganagar,Bhubaneswar	Class X (Central board of Secondary Education)	2017	9.00%

POSITIONS OF RESPONSIBILITY

Teaching Assistant, Control Systems & Flight Dynamics, IIT Hyderabad -Assisted in course delivery, supported students with technical concepts and problem-solving, and contributed to academic activities.

Represented IIT Hyderabad at the Inter-IIT Sports Meet in Football in 2025 and 2026, competing at the inter-institute level as a member of the institute team

SKILLS

Tools: ANSYS Fluent, Abaqus, SolidWorks, ICEM CFD, Simulink, 3D Printing

Programming: MATLAB, Python

Software: LaTeX, Adobe, Microsoft Office

RELEVANT COURSES

FEA :Finite Element Method ,**Strength/structures** :Flight Dynamics and Structures, **Aerospace fundamentals** :Aerodynamics and Propulsion, **Advanced aerodynamics:**High-Speed Aerodynamics, **Control:** Spacecraft Dynamics and Control, **Numerical methods:** Mathematical Methods for Engineers,**Programming/data/automation angle** : Machine Learning and its Applications

PROJECTS

Delivery and Surveillance Drone

- Designed and developed a quadcopter UAV for medical delivery, achieving a payload capacity of ~700 g and an operational range of 500 m. Contributed to **UAV design and numerical analysis** while leading the project team.

Role: Group Leader

Active Flow Control of Fluid-Structure Interaction in a Flexible Wing

- Investigating the use of **surface microjets to delay the onset of flutter** and studying their effect on the **fluid-structure interaction and aeroelastic response** of the wing

Effect of Ultrasound on Shear Wave Propagation in Hyperelastic Materials:

- Simulated the effect of ultrasonic excitation on shear wave propagation in hyperelastic materials using **Abaqus finite element analysis**

Autonomous Orbital Docking with a Tumbling Target

- Validated a **6-DOF Sliding Mode Control (SMC) architecture in Python** for autonomous orbital docking dynamics.
- Eliminated **zero-order-hold tracking lag by 100%** through analytical feedforward of target port angular rates.
- Implemented an **online RLS-RFF disturbance observer**, reducing actuator chatter by **56%**.

Flight Dynamics Project

- Performed a comprehensive study of aircraft longitudinal and lateral-directional dynamics using analytical and numerical methods; conducted **dynamic stability analysis of a MiG-21 aircraft in MATLAB**.

Design and Fabrication of Six-Bar Linkage

- Designed and fabricated a **six-bar slider-crank mechanism** for mining and oil & gas applications, contributing to the **CAD modelling and mechanical design** of the mechanism.

Experimental Investigation of Fluid-Structure Interaction (FSI) in a Flexible Mylar Sheet

- Conducted experiments to investigate **subcritical and supercritical bifurcation behaviour** in an oscillating Mylar sheet system at varying nozzle-to-sheet distances.
- Evaluated the system's dynamic response **with and without electromagnetic control** to investigate its influence on aeroelastic stability.

EXTRACURRICULAR

Achievements:

- **1st Prize**, Institute Football League, IIT Hyderabad
- **2nd Prize**, Chess Competition, MAE Department Day, IIT Hyderabad
- **2nd Prize**, Football Reliance Cup 2022 , Siksha O Anusandhan University
- **3rd Prize**, Inter-Hostel Football Competition, IIT Hyderabad
- **Represented IIT Hyderabad in Football** at the Inter-IIT Sports Meet in 2025 and 2026
- **Team Captain, Red Bull Campus Clutch East Zone Tournament** - Captained an independent Valorant team (fps) and competed in the East Zone tournament.

Hobbies: Football, Chess, Calisthenics, First-Person Shooter (FPS) Games

LANGUAGES

Proficient in **English**(Professional)**, Hindi**(Elementary) **and Odia**(Mother tongue)